

A Perturbative Approach to the Ground State of Interacting Fermions in the Mean Field Limit

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We begin by introducing the system in exam: a Fermi gas with a two-body interaction, composed of N particles

$$H_N = H_0 + \lambda \sum_{1 \leq i < j \leq N} V(x_i - x_j)$$

where

- H_0 is the kinetic energy;
- V is the two-body interaction potential:
 - (i) V is a Schwartz function;
 - (ii) V depends only on $|x_i - x_j|$;
 - (iii) $\hat{V}$ has compact support;
- λ is the coupling constant.

Let us recall the properties of the **free Fermi gas**. On the fermionic N -particle state space

$$\mathcal{H}_N = \left\{ \psi \in L^2(\mathbb{T}^{3N}) : \right. \\ \left. \psi(x_1, \dots, x_N) = (-1)^\sigma \psi(x_{\sigma(1)}, \dots, x_{\sigma(N)}) \right\}$$

the kinetic energy is the sum of kinetic energies, with a Slater determinant ground state

$$H_0 = \sum_{i=1}^N -\hbar^2 \Delta_i \quad \Psi_{\text{gs}} = \bigwedge_{k \in B_F} \psi_k$$

where $\psi_k(x) = (2\pi)^{-3/2} e^{-ik \cdot x}$.

Remark

$\hbar$ is chosen to be a physical parameter that we will tweak in the mean field limit. It is not the fundamental Planck's constant.

The Slater determinant is a consequence of Pauli's exclusion principle, manifest in the complete antisymmetry of the wave function Ψ_{gs} .

Momentum states ψ_k can be occupied by at most one particle.

The energy spectrum is $\epsilon_k = \hbar^2 |k|^2$: adding particles in the lowest energy states available fills a ball in momentum space. **FERMI BALL IMAGE**

The **Fermi momentum** is the radius of the highest occupied shell

$$k_F = \left(\frac{3}{4\pi} N \right)^{\frac{1}{3}} + \mathcal{O}(1)$$

$$B_F = \{k \in \mathbb{Z}^3 : |k| \leq k_F\}$$

Adding interactions introduces non-trivial correlations between particle states, and these properties are better understood from a statistical mechanics perspective: at thermal equilibrium the ground state energy is the zero temperature limit of the free energy.

Definition

A **thermal equilibrium state** at inverse temperature β is a quantum state $\langle \cdot \rangle_\beta \in \mathcal{B}(\mathcal{H}_N)^*$ that follows the Boltzmann distribution $p(\epsilon) = Z^{-1}e^{-\beta\epsilon}$:

$$\langle \cdot \rangle_\beta = \frac{\text{Tr}(\cdot e^{-\beta H_N})}{Z}, \quad Z = \text{Tr}(e^{-\beta H_N}).$$

$\langle \cdot \rangle_\beta$ is also called a *Gibbs state*.

The quantity Z is the **partition function** and it contains all the information we are looking for. The free energy is just

$$F = -\frac{\log(Z)}{\beta}, \quad E_{\text{gs}} = \lim_{\beta \rightarrow \infty} F$$

so we need to compute $\log(Z)$ first. To carry out such computation we work in the **grand canonical picture**, using the second quantization formalism. When working in the grand canonical picture, the energy acquires the additional term $-\mu\mathcal{N}$, with μ being the **chemical potential**, and $\mathcal{N}$ the number operator.

In second quantization, N is allowed to vary by modifying the state space to the **Fock space**. For fermions with one-particle state space $\mathfrak{h}$:

$$\mathcal{F}^F(\mathfrak{h}) = \bigoplus_{N=0}^{\infty} \bigwedge^N \mathfrak{h}$$

with the $N = 0$ sector being, conventionally, $\mathbb{C}$.

On a Fock space, observables are written in terms of **creation/annihilation** operators

$$\begin{aligned}a(f)\varphi_1 \otimes \cdots \otimes \varphi_N &= \langle f, \varphi_1 \rangle \varphi_2 \otimes \cdots \otimes \varphi_N \\ a^*(f)\varphi_1 \otimes \cdots \otimes \varphi_N &= f \otimes \varphi_1 \otimes \cdots \otimes \varphi_N.\end{aligned}$$

Their fermionic version is given by projecting onto the antisymmetric subspace of each N -particle sector:

$$a_-(f) = \sqrt{N}a(f) \quad a_-^*(f) = \sqrt{N}P_-a^*(f).$$

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These satisfy the **Canonical Anticommutation Relations**:

$$\{a_-(f), a_-^*(g)\} = \langle f, g \rangle \mathbf{I}$$

with all other anticommutators being zero.

With creation/annihilation operators, the Hamiltonian H_N is lifted to $\mathcal{F}^F(\mathfrak{h})$:

$$H = \bigoplus_{N=0}^{\infty} H_N$$

where

$$H_N = \sum_{k \in \mathbb{Z}^3} (\hbar^2 |k|^2 - \mu) a_k^* a_k + \lambda \sum_{p, q, k \in \mathbb{Z}^3} \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p.$$

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In the free Fermi gas, the Fermi momentum grows as $k_F = \mathcal{O}(N^{-1/3})$. Then the total energy in the ground state grows as $E_{\text{gs}}^{(0)} = \mathcal{O}(N^{5/3})$.

Adding the interaction means the potential energy will be of order N^2 as the number of pair-wise interactions is $N(N-1)/2$.

Mean field scaling

To get a non-trivial limit for the true ground state energy as $N \rightarrow \infty$, we define

$$\hbar = N^{-1/3} \quad \text{and} \quad \lambda = N^{-1}.$$

In this mean field scaling limit, exact methods show that the ground state energy can be approximated by

$$E_{\text{gs}} = E_N^{\text{HF}} + E_N^{\text{RPA}} + \mathcal{O}(N^{-\frac{1}{3}-\alpha}).$$

Here

- $E^{\text{HF}} = \mathcal{O}(N) + \mathcal{O}(1)$ is the contribution from the **Hartree-Fock (HF)** approximation;
- $E^{\text{RPA}} = \mathcal{O}(N^{-1/3})$ is the contribution from the **Random Phase Approximation (RPA)**;
- $\alpha > 0$ indicates that further contributions will be subleading with respect to the RPA.

Our work introduces rigorous perturbation theory into the picture. We start by the first order, and then lay the groundwork for the second.

- The first order approximation leads to the HF term, containing the unperturbed energy, the **direct** ($\mathcal{O}(N)$) and the **exchange** ($\mathcal{O}(1)$) corrections;
- The RPA contains higher order direct contributions (all expected to be of order $\mathcal{O}(N^{-1/3})$);
- Already at second order, the exchange contribution is expected to be subleading with respect to the RPA, and thus fall in the error term.

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Because H is the sum of a kinetic term and an interaction term, the partition function cannot be computed exactly, but we can formally expand it in powers of λ :

Duhamel's formula

The following approximation for $\log(Z)$ holds formally at second order in λ :

$$\begin{aligned}\log \operatorname{Tr} e^{-\beta H} &= \\ &= \log \operatorname{Tr} e^{-\beta H_0} - \lambda \beta \langle V \rangle_\beta \\ &\quad + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' (\langle V(\tau') V(\tau) \rangle_\beta - \langle V(\tau') \rangle_\beta \langle V(\tau) \rangle_\beta)\end{aligned}$$

In this formula:

- The expectation $\langle \cdot \rangle_\beta$ is done with respect to the kinetic term H_0 ;
- $V(\tau) = e^{-\tau H_0} V e^{\tau H_0}$ is the “evolution” of V at inverse temperature τ ;
- $Z_0 = \log \text{Tr } e^{-\beta H_0}$ is just the partition function for the free system
 $\implies$ terms in λ and λ^2 are higher order corrections to Z_0 .

The expectations $\langle V \rangle_\beta$ and $\langle V(\tau') V(\tau) \rangle_\beta - \langle V(\tau') \rangle_\beta \langle V(\tau) \rangle_\beta$ can be computed using **Wick's theorem**.

Wick's theorem

If $H = \sum_k \epsilon_k a_k^* a_k$ with $\sum_k e^{-\beta \epsilon_k} < \infty$, then, given a family of creation and annihilation operators $A_1, \dots, A_{2n}$, it holds that

$$\langle A_1 \cdots A_{2n} \rangle_\beta = \sum_{\sigma \in P_{2n}} (-1)^\sigma \prod_{j=1}^{2n-1} \langle A_{\sigma(j)} A_{\sigma(j+1)} \rangle_\beta$$
$$\langle A_1 \cdots A_{2n-1} \rangle_\beta = 0$$

where P_{2n} is the set of **pairings** of $2n$ elements.

Remark

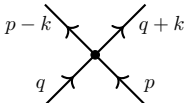
Wick's theorem is the defining property of **quasi-free states**.
Gibbs states for quadratic Hamiltonians are quasi-free.

Wick's theorem can be used diagrammatically:

- to a two-point function $\langle a_k^*(\tau) a_{k'}(\tau') \rangle_\beta$ is associated the **propagator** $G_{k,k'}(\tau, \tau')$:

$$G_{k,k'}(\tau, \tau') = \tau, k \longrightarrow \tau', k'$$

- to the interaction term is associated a **vertex**:

$$\sum_{p,q,k} \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p =$$


- propagators connect vertices to form a **Feynman diagram**.

Linked cluster theorem

For a product of n vertices $V_1, \dots, V_n$, we have the following identity in terms of values of **connected Feynman diagrams**:

$$\langle V_1 \cdots V_n \rangle_\beta - \langle V_1 \rangle_\beta \cdots \langle V_n \rangle_\beta = \sum_{\Gamma \in \mathcal{G}_{n,C}} \text{val}(\Gamma),$$

where $\mathcal{G}_{n,C}$ is the set of all connected Feynman diagrams that contain n vertices.

Looking at the second order expansion of $\log(Z)$:

- At first order: $\langle V \rangle_\beta$ is the sum of all connected Feynman diagrams with *one vertex*;
- At second order: $\langle V(\tau')V(\tau) \rangle_\beta - \langle V(\tau') \rangle_\beta \langle V(\tau) \rangle_\beta$ is the sum of all connected Feynman diagrams with *two vertices*.

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First, we explicitly evaluate all Feynman diagrams that appear at first order:

$$\begin{aligned}
 \langle V \rangle_\beta &= \text{[Feynman diagram 1]} + \text{[Feynman diagram 2]} \\
 &= \sum_{p,k \in \mathbb{Z}^3} \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k}.
 \end{aligned}$$

The Feynman diagrams represent the first-order contributions to the expectation value of the potential energy. Each diagram consists of two vertices connected by two fermion lines, forming a figure-eight shape. The first diagram has a direct interaction, and the second diagram has an exchange interaction.

Here, the $\hat{V}(0)$ is the order direct term, while $\hat{V}(k)$ is the exchange term.

Lemma

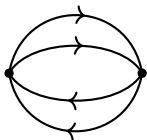
The series $\sum_{p,k} (\hat{V}(0) - \hat{V}(k)) n_p n_{p-k}$ converges uniformly in β (away from zero).

Theorem: mean field scaling of E_N^{HF}

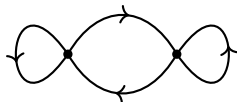
The ground state energy of the interacting Fermi gas in the mean field limit exhibits the following scaling in N :

$$\begin{aligned} E_N^{\text{HF}} &= \lim_{\beta \rightarrow \infty} \left(-\frac{\log(Z_0) - \lambda \beta \langle V \rangle_\beta}{\beta} \right) \\ &= E_N^{(0)} + \sum_{k \in \mathbb{Z}^3} \sum_{\substack{p \in B_F: \\ p-k \in B_F}} (\hat{V}(0) - \hat{V}(k)) \\ &= \mathcal{O}(N) + \mathcal{O}(1). \end{aligned}$$

At second order, the connected Feynman diagrams are split into two possible topologies:



No tadpoles



Two tadpoles

We expect the first kind of diagrams to represent the direct and the exchange contributions, while the tadpole diagrams to contribute exactly zero.

However, we set our calculations in formal perturbation theory. In the fermionic case this involves using **Grassmann variables** and **Grassmann integrals** to reproduce Wick's theorem.

In $\langle V(\tau)V(\tau') \rangle_\beta$ the vertices are at different temperatures: in formal perturbation theory, one should introduce τ -ordering to reproduce Duhamel's formula.

Remark

It may be that the correct perturbations at orders ≥ 2 cannot be obtained without looking deeper into the identification between the operatorial and Grassmannian perturbative series.

Indeed, if we evaluate tadpole diagrams we obtain **divergent contributions** after applying Duhamel's formula.

Example: tadpole divergence

$$\lim_{\beta \rightarrow \infty} \frac{\beta}{2} \left(\text{Diagram} \right) = \lim_{\beta \rightarrow \infty} \left(-\frac{\beta}{2} \sum_{p,q,k} \hat{V}(0) n_q^2 n_p n_k \right) = -\infty$$

Furthermore, diagrams that contribute to the RPA term break the expected scaling in N .

Example: second order direct contribution

$$\begin{aligned}
 E_{\text{D}}^{(2)} &= -2\lambda^2 \sum_{k \in \mathbb{Z}^3} \sum_{\substack{p, q \in B_{\text{F}}: \\ q+k, p-k \in B_{\text{F}}}} \frac{\hat{V}(k)^2}{\hbar^2((p-k)^2 + (q+k)^2 - q^2 - p^2)} \\
 &= \mathcal{O}(N^{1/3}).
 \end{aligned}$$